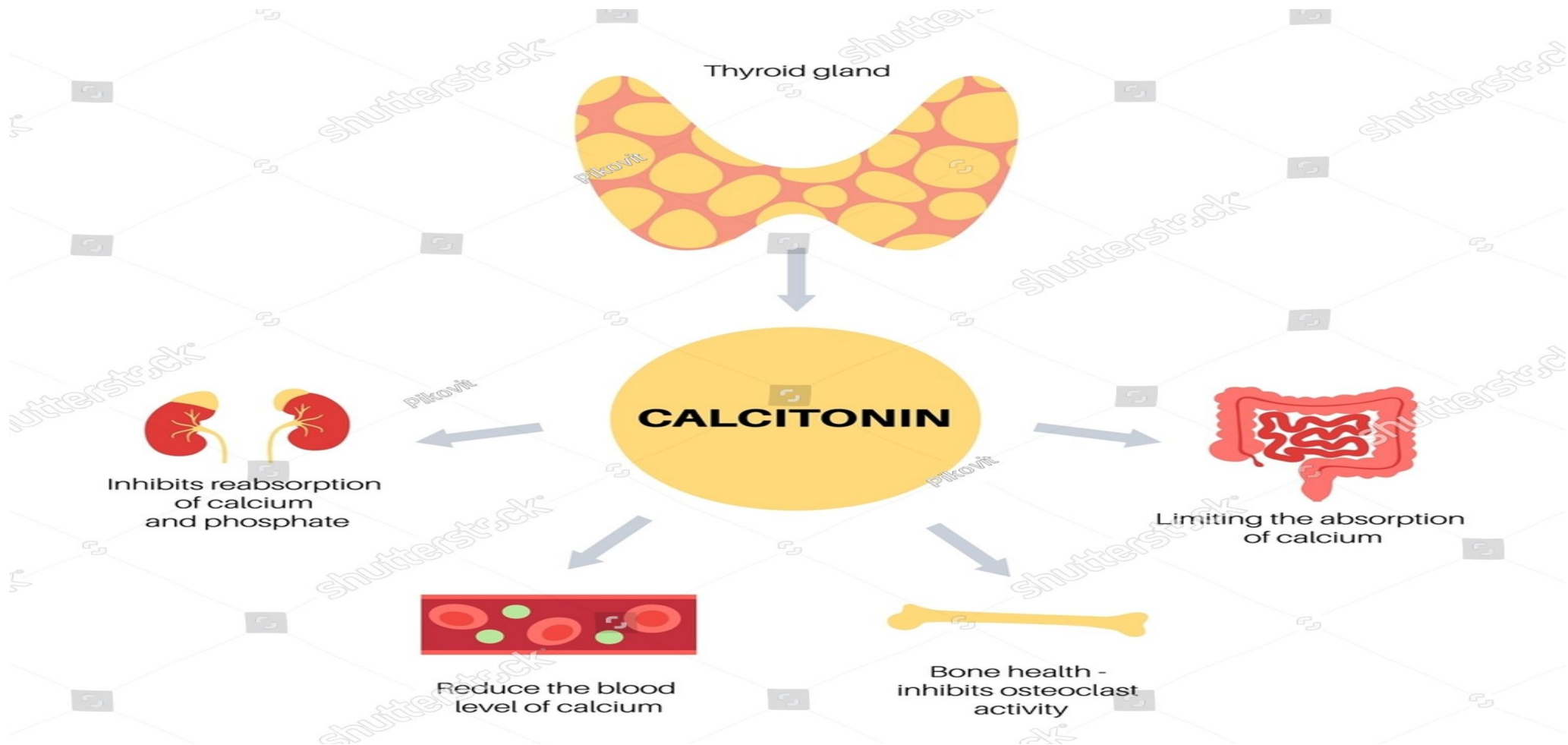


Parathyroid hormone, Calcitonin and Vitamin D



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Objectives:

Describe roles of PTH, calcitonin and vitamin D

Explain regulation of calcium levels

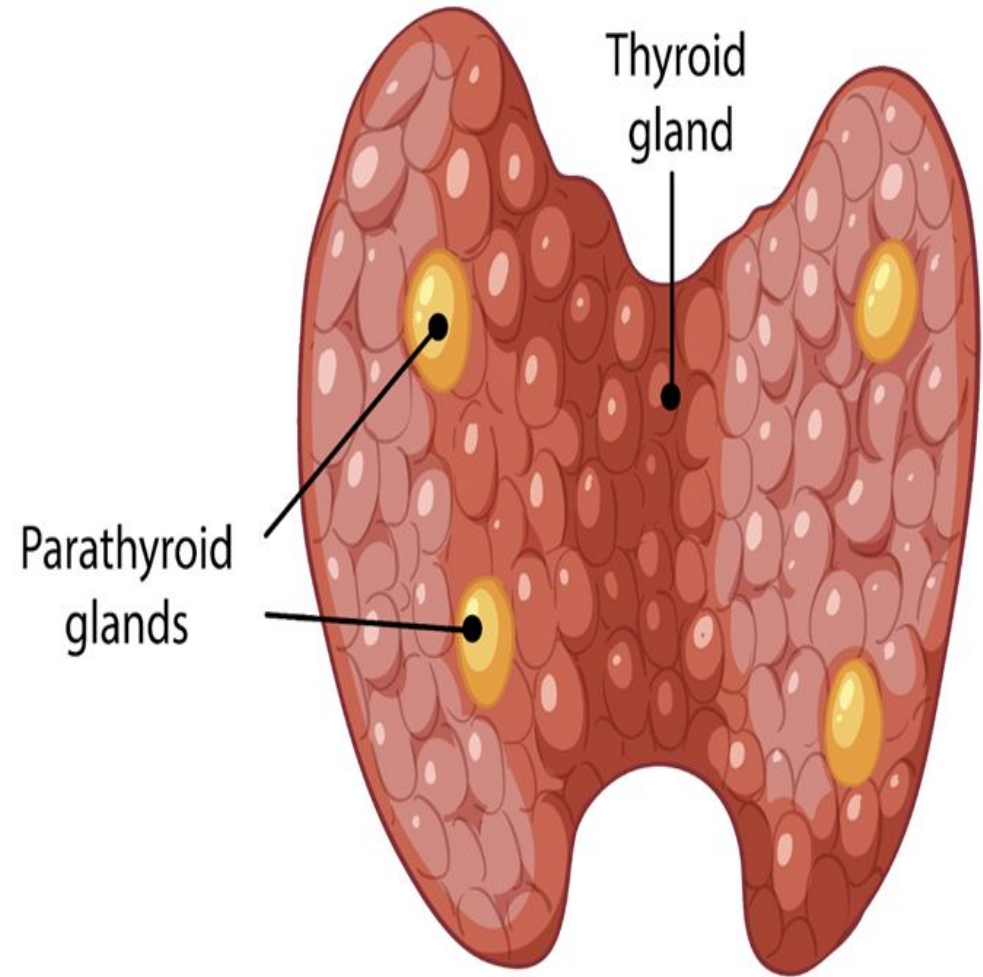
Relate hormones to bone metabolism

INTRODUCTION

- There are four parathyroid glands, which are situated on the posterior surface of upper and lower poles of thyroid gland.
- Parathyroid glands are very small in size. Each gland is about the size of a pea or a grain of rice, measuring about 6 mm long, 3 mm wide and 2 mm thick, with dark brown color.

Histology:

Parathyroid gland is made up of chief cells and oxyphil cells.



Function of the Parathyroid Glands

Parathyroid hormone performs following functions in the body.

□ Produce PTH:

They make and release parathyroid hormone (PTH), also known as Parathormone.

□ Control Calcium:

PTH regulates and controls the amount of **calcium** in the blood and bones.

□ Balance Phosphate:

It also helps manage **phosphate** levels in the body.

Function of the parathyroid glands

- Parathyroid glands produce parathyroid hormone (PTH), which plays a key role in the regulation of calcium levels in the blood.

Precise calcium levels are important in the human body, since small changes can cause muscle and nerve problems.

The parathyroid hormone stimulates the following functions:

- Release of calcium by bones into the bloodstream
- Absorption of calcium from food by the intestines
- Conservation of calcium by the kidneys
- Stimulates cells in the kidney to transform weaker forms of vitamin D into the form that is strongest at absorbing calcium from the intestines
- Balance Phosphate:

It also helps manage **phosphate** levels in the body.

How PTH Works

Bones:

PTH tells bones to release stored calcium into the bloodstream.

Intestines:

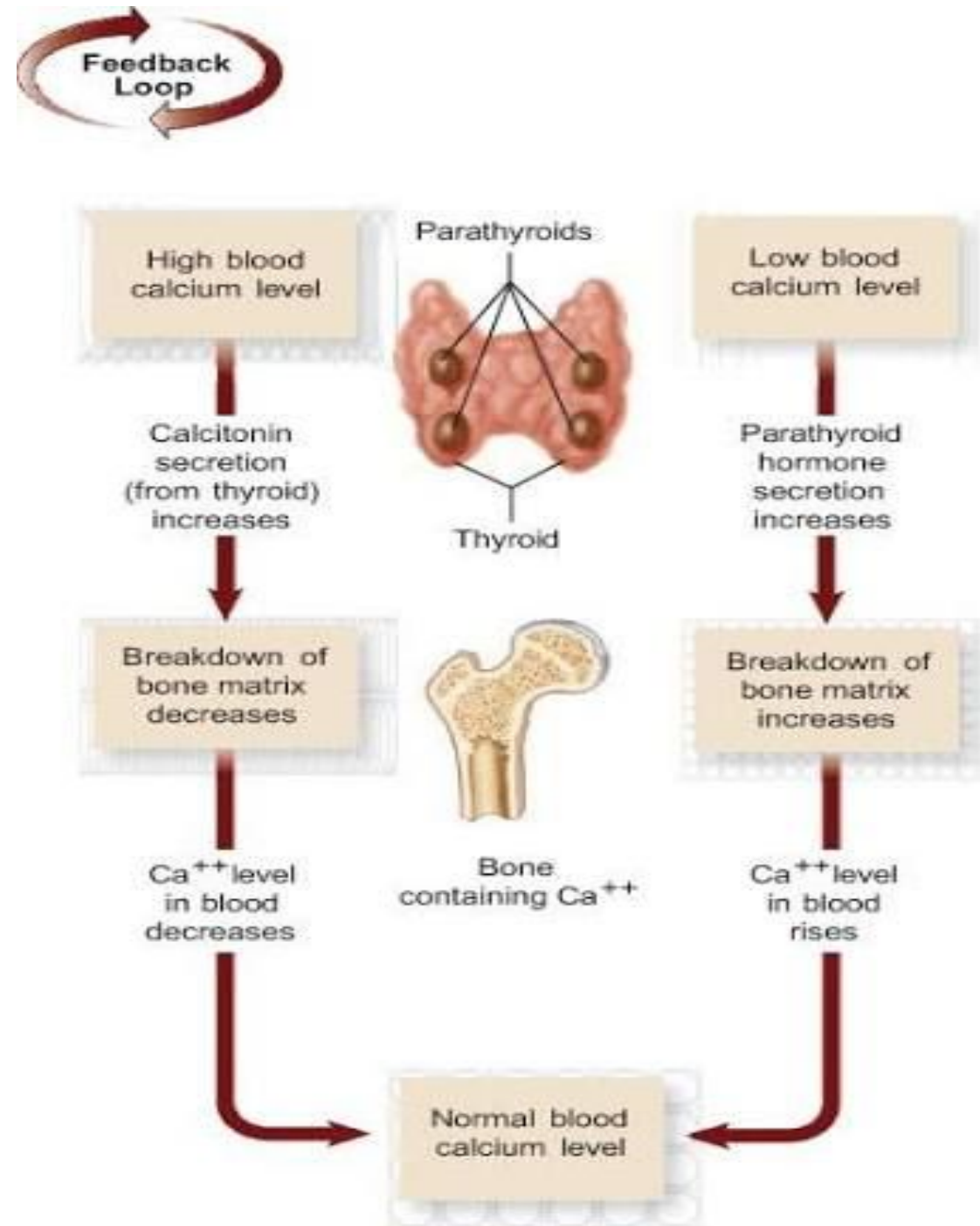
Parathyroid hormone indirectly increases calcium absorption from food in the intestine. It helps the gut absorb more calcium from the food you eat.

Kidneys:

Parathyroid hormone reduces loss of calcium in urine. Parathyroid hormone also stimulates the production of active vitamin D in the kidneys.

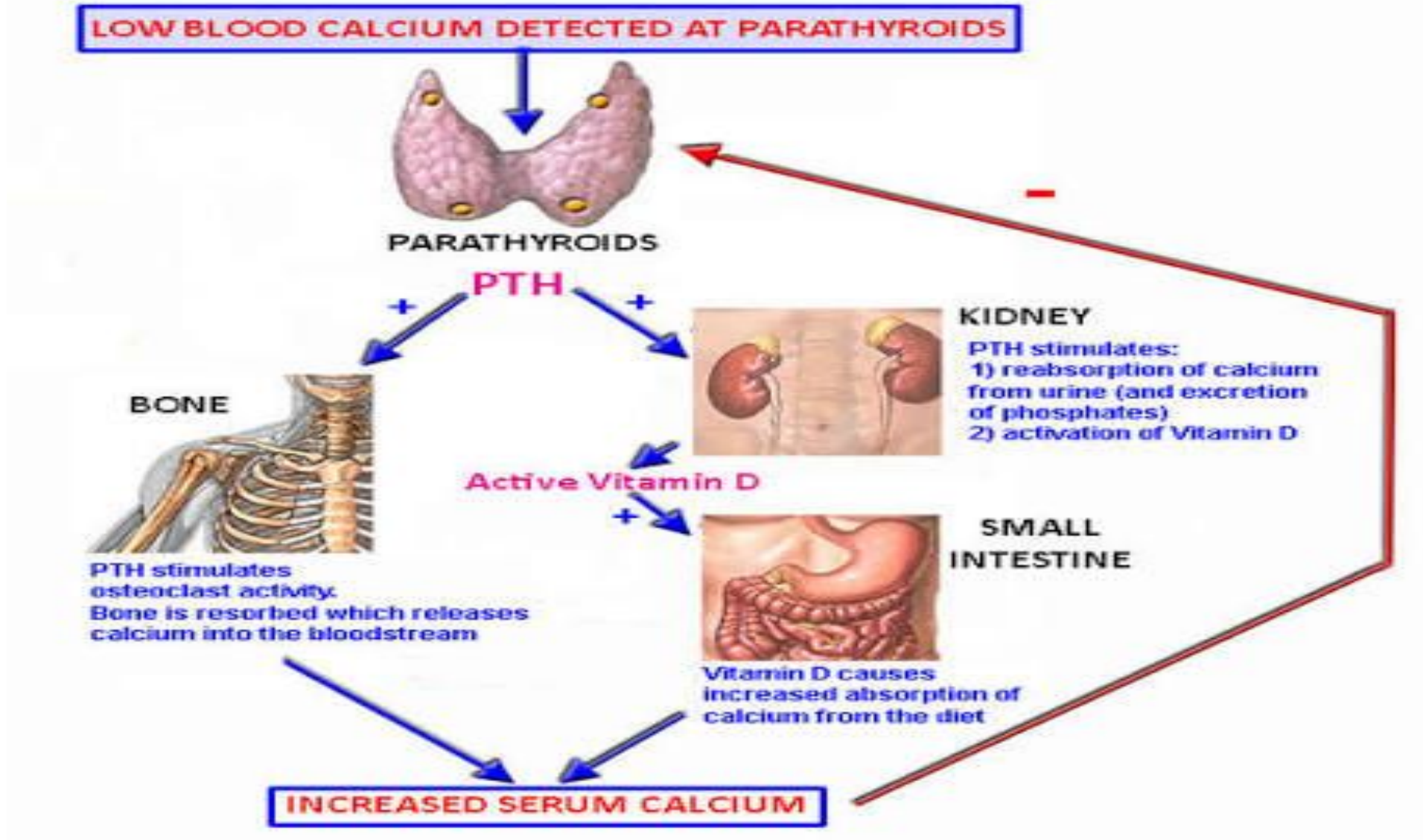
Feedback Loop:

When blood calcium drops, the glands make more PTH; when calcium rises, they make less.



How PTH Works:

It tells bones to release calcium, tells kidneys to save calcium, and helps intestines absorb more calcium from food



Conditions related to parathyroid hormone issues

When your parathyroid hormone levels are off balance, it causes certain health conditions.

1-High PTH levels (**hyperparathyroidism**)

2-Low PTH levels (**Hypoparathyroidism**)

Hyperparathyroidism

A high parathyroid hormone (PTH) level causes your body to pull too much calcium from your bones and absorb more through your gut, leading to a condition called hyperparathyroidism.

High PTH levels (hyperparathyroidism) cause:

- High blood calcium levels (hypercalcemia)
- Low levels of phosphorus in your blood (hypophosphatemia)
- Hyperparathyroidism has several possible causes. Some causes include parathyroid cancer, parathyroid adenoma and kidney disease.

Common Symptoms

Constant fatigue and muscle weakness

Frequent urination and extreme thirst

Bone and joint pain

Digestive issues like constipation, nausea, or belly pain

Mood changes such as depression, anxiety, or trouble concentrating

Long-Term Complications

Osteoporosis: Bones lose calcium, making them weak and easy to break.

Kidney stones: Excess calcium builds up in your urine and forms hard deposits.

Heart problems: High blood calcium links to high blood pressure and irregular heartbeats.

HYPERPARATHYROIDISM

SYMPTOMS



**Poor appetite,
constipation, & other
digestive symptoms**



**High blood
pressure**



**Fatigue and
low mood**



**Increased
thirst/urination**



Kidney stones



**Bone &
joint pain**

Hypoparathyroidism

Hypoparathyroidism is a rare condition that happens when you have low levels of parathyroid hormone in your blood.

This causes low levels of calcium (**hypocalcemia**) and high levels of phosphorous (**hyperphosphatemia**) in blood. Approximately 75% of hypoparathyroidism cases are from accidental damage to parathyroid glands from neck or thyroid surgery. There are other causes, too, like certain genetic and autoimmune conditions.

If the cause isn't due to surgery, in most cases, hypoparathyroidism progresses very gradually and symptoms can be mild. Symptoms include:

- Tingling in your lips, fingers and toes.
- “Brain fog” or confusion.
- Muscle cramps, Muscle spasms.
- Abdominal pain.
- Abnormal heart rhythm (arrhythmia).
- Brittle nails, Dry hair and dry scaly skin.
- Cataracts.
- Weakened tooth enamel (in children).

Some symptoms of hypoparathyroidism



Muscle cramps and spasms



“Brain fog” or confusion



Abnormal heart rhythm



Tingling or numbness in your lips, fingers and toes



Brittle nails and dry hair or skin

Diagnosis of parathyroid disorders

Diagnosing parathyroid problems depends on the condition, but may include:

Hyperparathyroidism

Blood tests to check for calcium, phosphorus, magnesium and PTH levels; bone x-rays; kidney scans.

A small tag of tissue may be removed and analyzed (biopsy).

Hypoparathyroidism

Blood tests to check for calcium, phosphorus, magnesium and PTH levels; tests (including ECG) to check for heart arrhythmia (irregular heart beat).

Multiple endocrine neoplasia 1

Blood tests; scans, including MRI and CT scans.

Treatment for Parathyroid disorders

Treatment depends on the condition and its severity, but may include:

Hyperparathyroidism –

In severe cases, the parathyroid glands and associated tumours may need to be surgically removed.

Hypoparathyroidism –

lifelong vitamin D and calcium carbonate supplements, and a high calcium, low phosphorus diet. An acute attack of hypoparathyroidism must be treated in hospital with intravenous calcium.

Multiple endocrine neoplasia 1 - radiation therapy and surgery to remove the tumours and associated glands. After surgery, hormone replacement therapy is needed.

Complications of Parathyroid disorders

Complications caused by untreated parathyroid disorders can include:

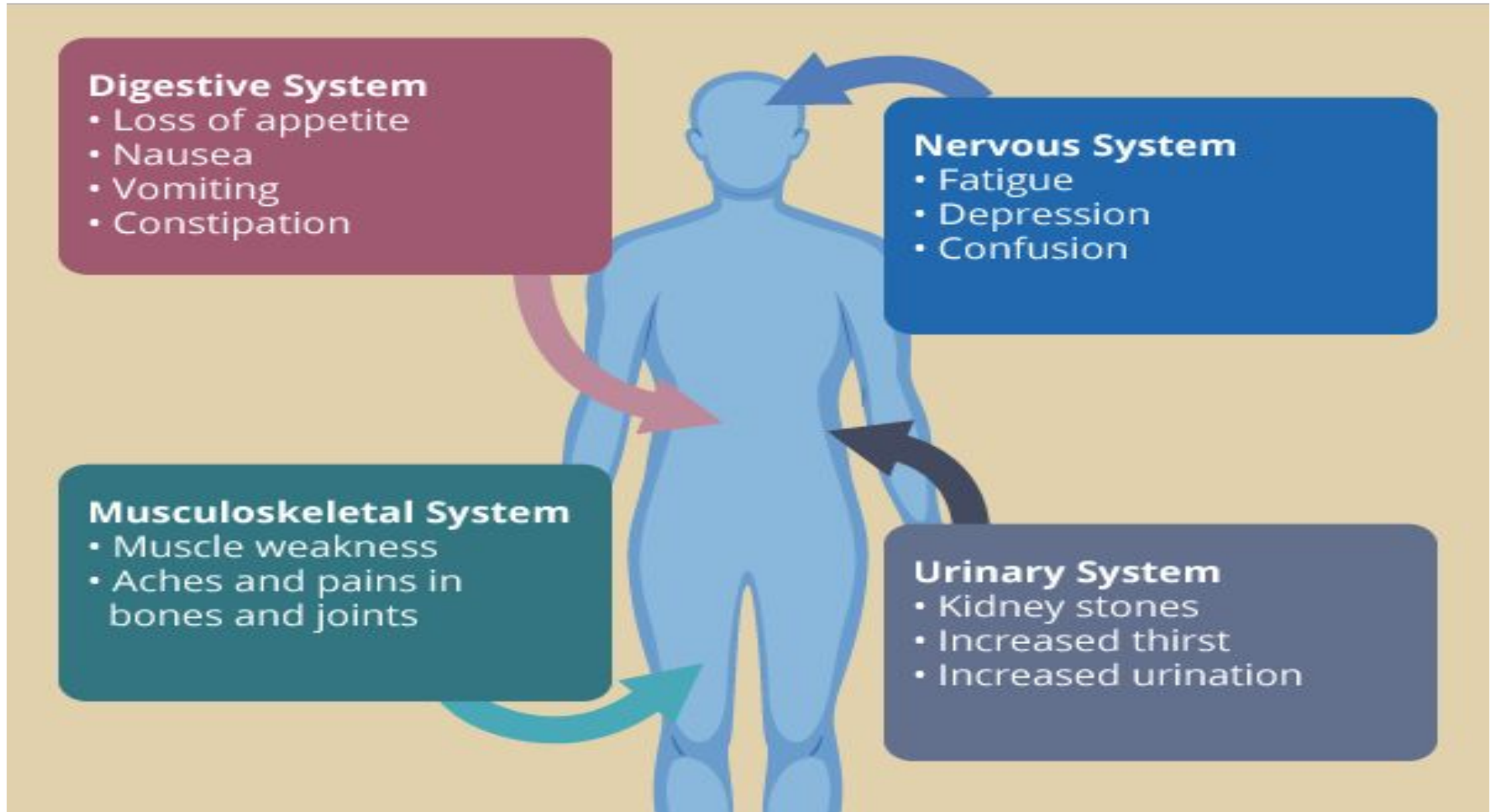
Hyperparathyroidism –

kidney stones,
urinary tract infections,
pancreatitis (inflammation of the pancreas) and
damage to bones.

Hypoparathyroidism –

Teeth fail to form or are slow to emerge,
Impaired mental development,
Addison's disease and
Dysmenorrhea (painful menstrual periods).

Primary Hyperparathyroidism



Hyperparathyroidism

Hyperthyroidism

Parathyroid glands
affected



Thyroid gland
affected



Hormones involved

PTH

T3, T4

Symptoms associated



High calcium



Bone & joint
pain



Kidney stones



Memory
problems



Rapid heartbeat



Unexplained
weight loss



Heat intolerance



Tremors

Common treatment

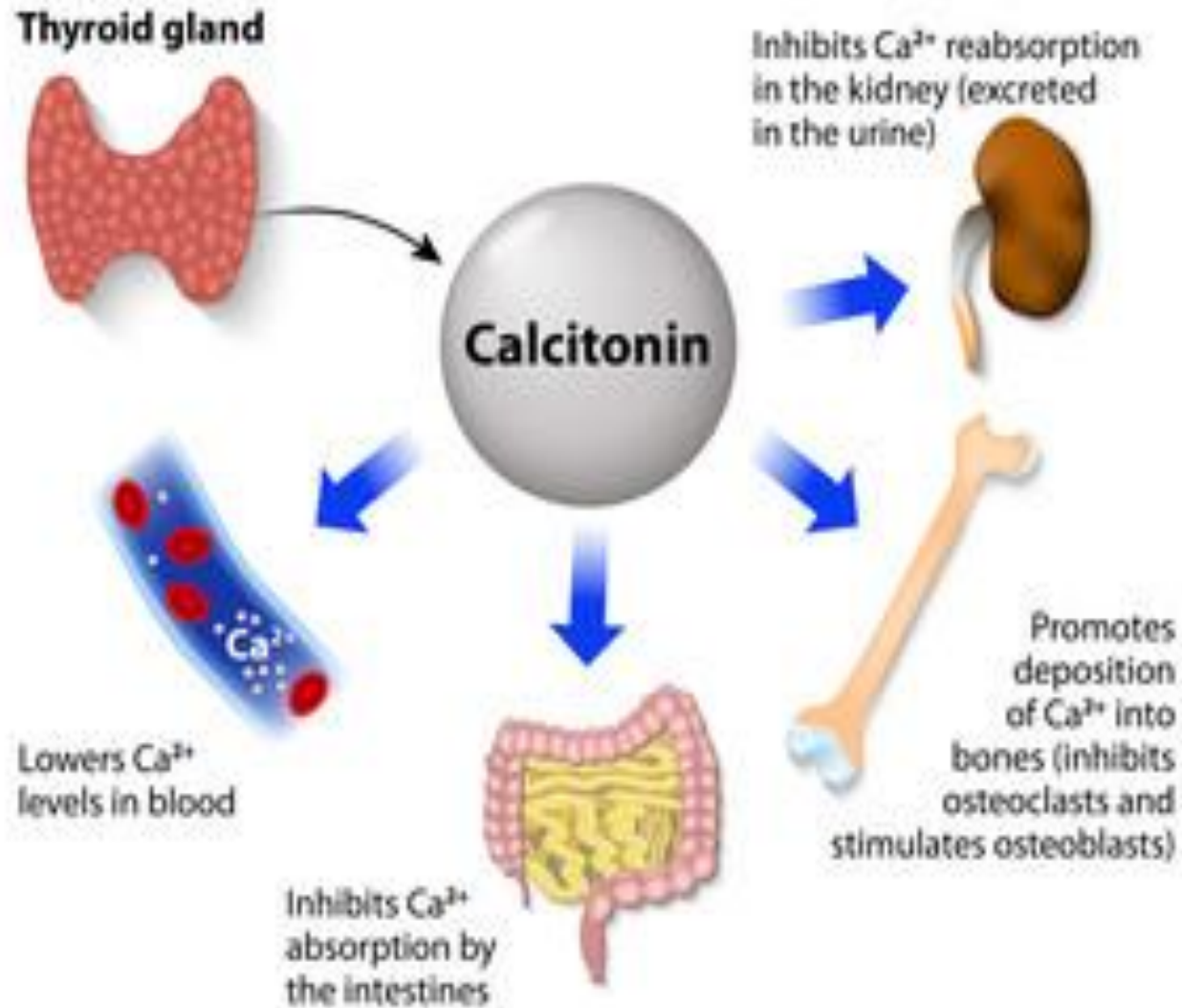


Minimally invasive
surgery



Medication or
radioactive iodine

Calcitonin and Vitamin – D



Vitamin D Foods



Calcitonin

CALCITONIN

It is a peptide hormone secreted by the thyroid gland, tends to decrease plasma calcium concentration and, in general, has effects opposite to those of PTH.

Synthesis and secretion of calcitonin occur in the parafollicular cells, or C cells, lying in the interstitial fluid between the follicles of the thyroid gland.

The primary stimulus for calcitonin secretion is increased extracellular fluid calcium ion concentration.

In contrast, PTH secretion is stimulated by decreased calcium concentration.

Regulation of Calcium Secretions

- Increased Plasma Calcium Concentration Stimulates Calcitonin Secretion.
 - The primary stimulus for calcitonin secretion is increased extracellular fluid calcium ion concentration.
 - In contrast, PTH secretion is stimulated by decreased calcium concentration.
 - High calcium content in Plasma stimulate the calcitonin secretion through a calcium receptor in parafollicular cells
- Gastrin is also known to stimulate the release of calcitonin.
- Calcitonin Decreases Plasma Calcium Concentration.

1. If the Calcium levels in the blood rise above a set point.

2. The Thyroid Gland releases Calcintonin

Calcitonin

3. Blood Calcium levels falls

High
Blood Calcium Level Regulation

Low

4. If the Calcium levels fall below a set point...

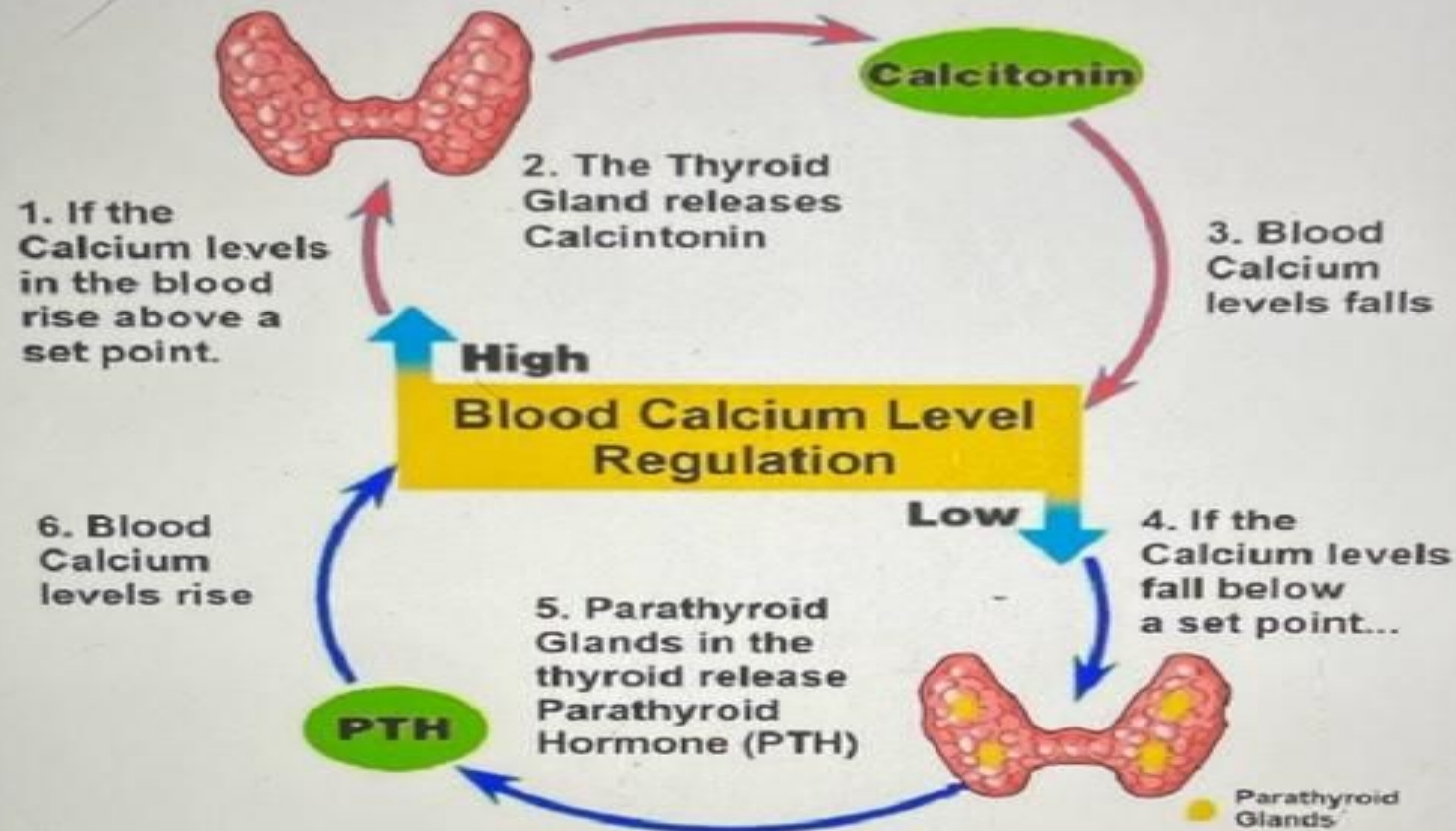
5. Parathyroid Glands in the thyroid release Parathyroid Hormone (PTH)

PTH

6. Blood Calcium levels rise

Parathyroid Glands

Based on the following diagram, which of the following will happen if the blood Calcium level increases?



- ☐ Bones are broken down to release Ca^{++} ions
- ☐ The Ca^{++} are absorbed from the digestive system
- ☐ The parathyroid glands release the parathyroid hormone
- ☐ The thyroid gland releases calcitonin

Chemistry and Synthesis

Calcitonin is a Polypeptide chain with 32 amino acids.

Its molecular weight is about 3400

It is synthesized from Procalcitonin.

Plasma Level of calcitonin is 1 to 2 ng/dL

It has half life of 5-10 minutes

Metabolism:

Calcitonin is degraded and excreted by liver and kidney

Actions of Calcitonin

Calcitonin reduces the blood calcium level by acting on bones, kidneys and intestine.

1-On bones

Calcitonin inhibits the reabsorption of phosphate from bone and stimulates the deposition of phosphate on bones.

2-On kidney

Calcitonin increases excretion of calcium through urine, by inhibiting the reabsorption from the renal tubules.

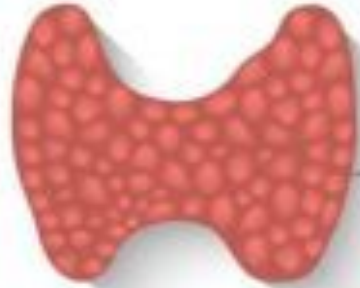
3-On intestine

Calcitonin prevents the absorption of calcium from intestine into the blood.

4- On Blood

It decreases the blood level of phosphate by acting on bones and kidneys

Thyroid gland

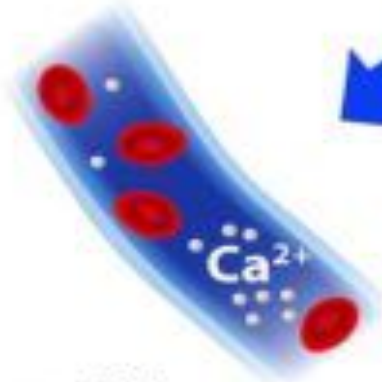


Calcitonin

Inhibits Ca^{2+} reabsorption
in the kidney (excreted
in the urine)



Lowers Ca^{2+}
levels in blood



Inhibits Ca^{2+}
absorption by
the intestines



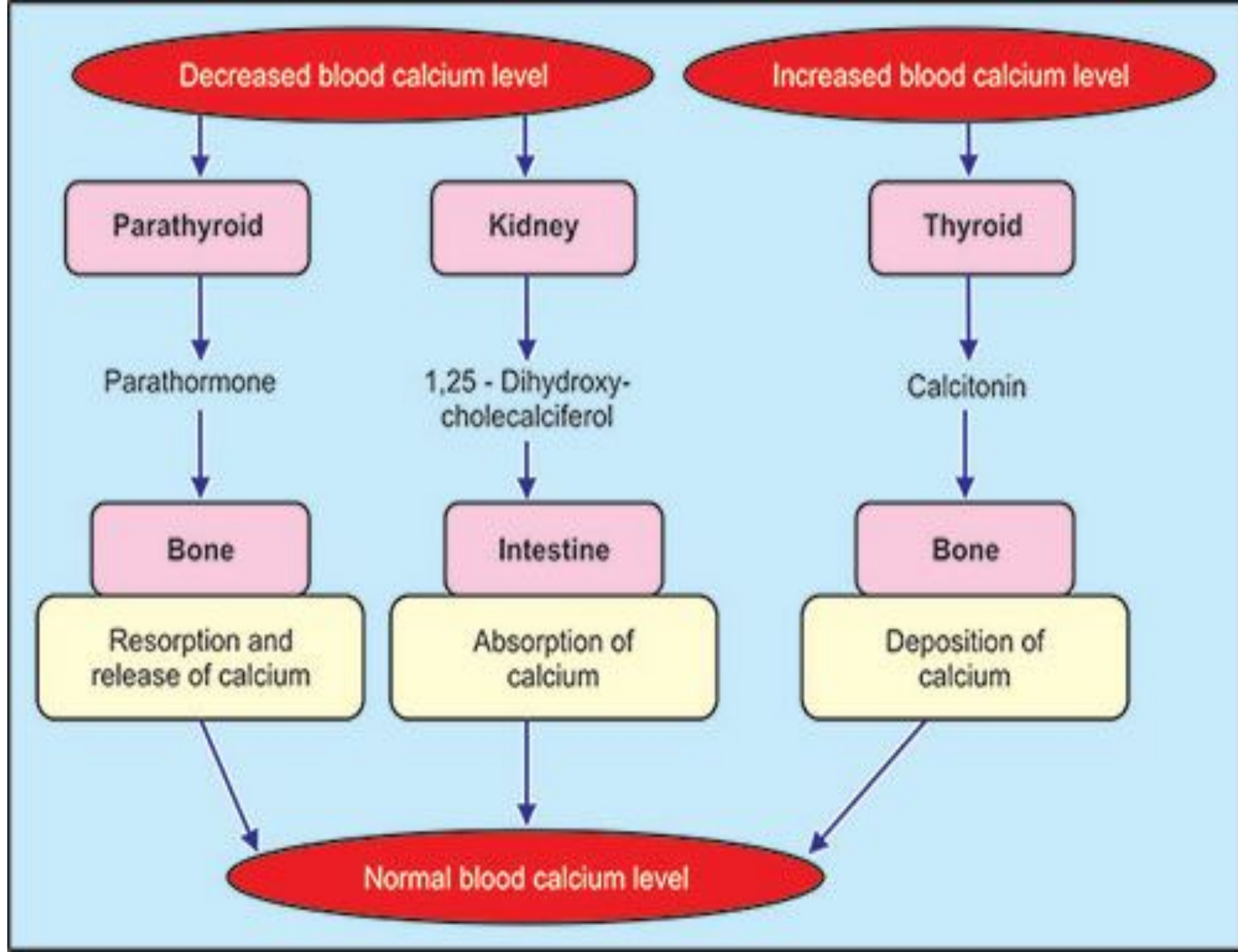
Promotes
deposition
of Ca^{2+} into
bones (inhibits
osteoclasts and
stimulates osteoblasts)



REGULATION OF CALCITONIN SECRETION

High calcium content in plasma stimulates the calcitonin secretion through a calcium receptor in parafollicular cells.

- Increased Plasma Calcium Concentration Stimulates Calcitonin Secretion.
- Calcitonin Decreases Plasma Calcium Concentration.
- Plays an important role in controlling the blood calcium level.
- It decreases the blood calcium level and thereby counteracts parathormone.
- Calcitonin reduces the blood calcium level by acting on:
 - Bones
 - Kidneys
 - Intestine



Vitamin D: Production, Metabolism and Mechanisms of Action

VITAMIN D

FOODS, FRUITS & VEGETABLES



VITAMIN-D

Vitamin D production in the skin under the influence of sunlight (UVB) is maximized at levels of sunlight exposure that do not burn the skin.

Further metabolism of vitamin D to its major circulating form (25(OH)D) and hormonal form (1,25(OH)₂D) takes place in the liver and kidney, respectively

Vitamin D is both a nutrient we eat and a hormone our bodies make.

It is a fat-soluble vitamin that has long been known to help the body absorb and retain Calcium and Phosphorus; both are critical for building bone.

Sources of vitamin D

We can often get the majority of the vitamin D intake from sunlight exposure. We cannot solely rely on sunlight exposure for vitamin D production. During the winter months, when the sun is not as strong, everyone can benefit from vitamin D supplements.

The following foods are a source of vitamin D:

- fatty fish, such as salmon, mackerel, and tuna
- egg yolks
- cheese
- beef liver
- mushrooms
- fortified milk
- fortified cereals and juices

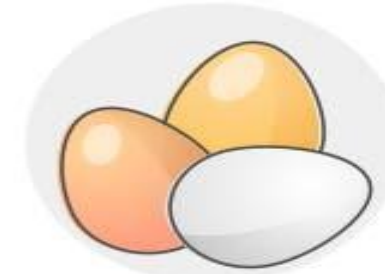
SOURCES OF VITAMIN D



MILK



SUN LIGHT



EGGS



MUSHROOMS



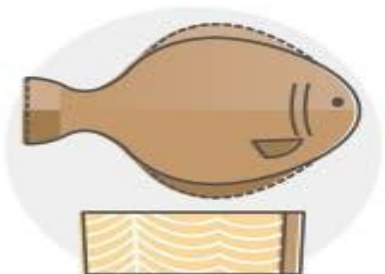
PORK



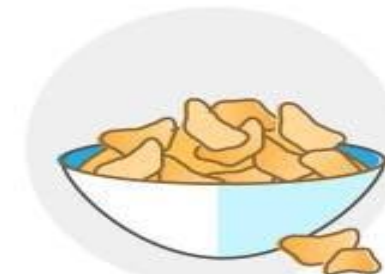
BEEF LIVER



RICOTTA CHEESE



SOLE OR FLOUNDER



CEREAL

ACTIONS OF VITAMIN D

The active form of vitamin D, 1,25-dihydroxycholecalciferol, has several effects on the intestines, kidneys, and bones that increase absorption of calcium and phosphate into the extracellular fluid and contribute to feedback regulation of these substances.

An adequate intake of vitamin D may support good immune function and reduce the risk of autoimmune diseases.

Vitamin D promotes calcium absorption in the gut and maintains adequate serum calcium and phosphate concentrations to enable normal bone mineralization and to prevent hypocalcemic tetany (involuntary contraction of muscles, leading to cramps and spasms).

Vitamin D promotes Phosphate Absorption by the intestines.

Vitamin D decreases Renal Calcium and Phosphate Excretion.

Vitamin D plays important roles in bone resorption and bone deposition.

Functions of Vitamin D

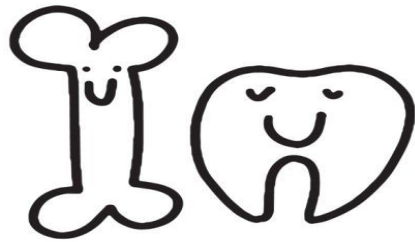


ACTIONS OF VITAMIN D

- It is also needed for bone growth and bone remodeling by osteoblasts and osteoclasts .
- Without sufficient vitamin D, bones can become thin, brittle, or misshapen.
- Vitamin D sufficiency prevents rickets in children and osteomalacia in adults.
- Long-term vitamin D deficiency can also present as osteoporosis.
- Together with calcium, vitamin D also helps protect older adults from osteoporosis.

VITAMIN D

BENEFITS



CONTRIBUTES
TO BONES
HEALTH



FACILITATES
HORMONE
REGULATION



HELPS
MANAGE BLOOD
SUGAR LEVELS

ENHANCES
THE IMMUNE
SYSTEM



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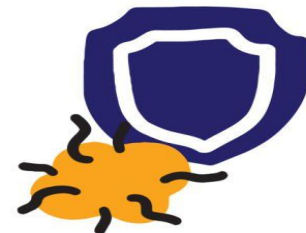


HELPS WITH
CONCENTRATION
LEARNING AND
MEMORY

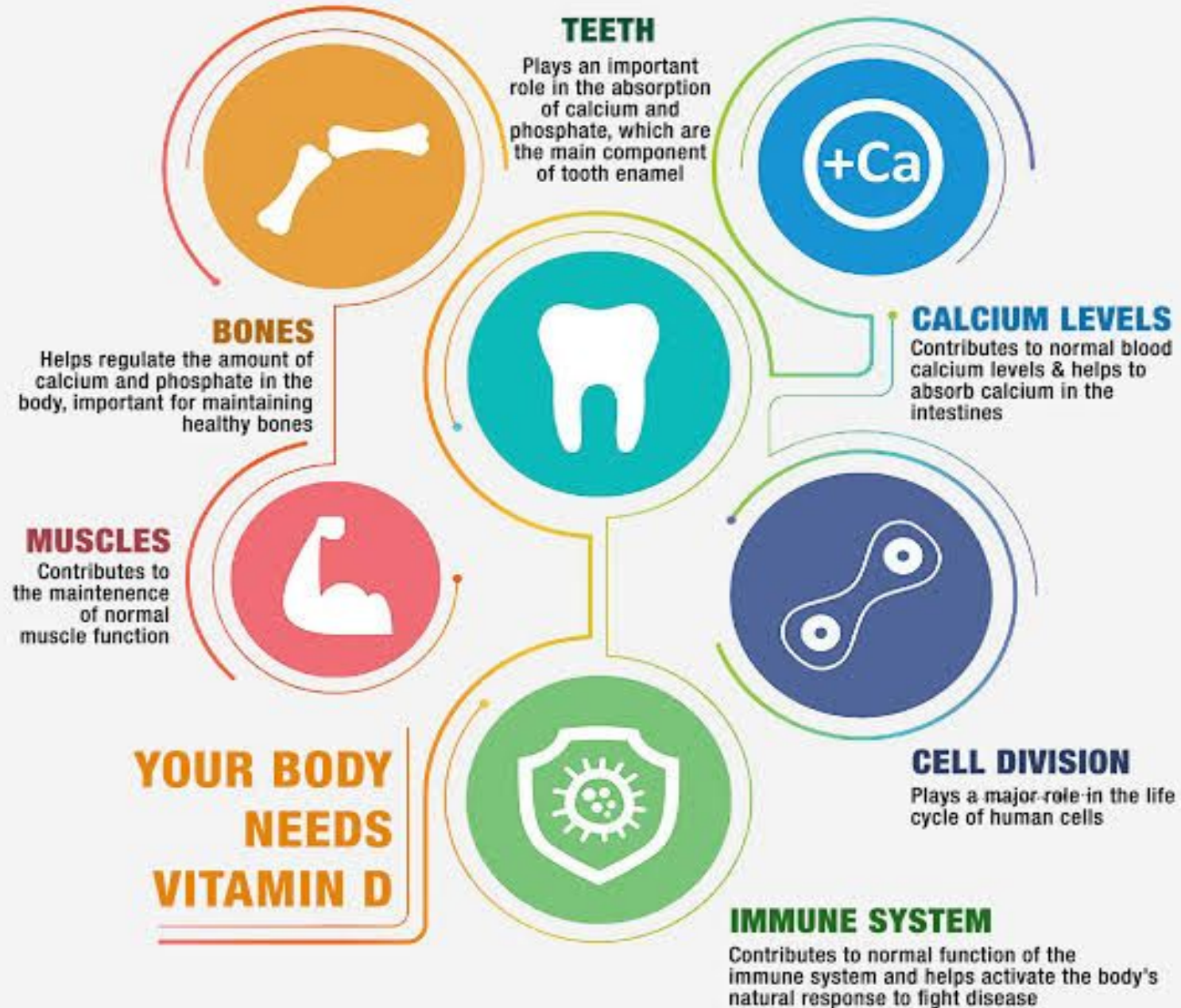


IMPROVES
HEART HEALTH

PROTECT
AGAINST
CANCER



WHY VITAMIN D?



SYMPTOMS OF VITAMIN D DEFICIENCY



FATIGUE



JOINT PAIN



HAIR LOSS



WEIGHT GAIN



DEPRESSION



MOOD CHANGES



BONE LOSS



MUSCLES CRAMP



FREQUENT ILLNESS

Vitamin D Deficiency

Vitamin D deficiency may occur from a lack in the diet, poor absorption, or having a metabolic need for higher amounts.

If one is not eating enough vitamin D and does not receive enough ultraviolet sun exposure over an extended period a deficiency may arise.

People who cannot tolerate or do not eat milk, eggs, and fish, such as those with a lactose intolerance or who follow a vegan diet, are at higher risk for a deficiency.

Other people at high risk of vitamin D deficiency include:

- ❑ People with inflammatory bowel disease
- ❑ People who are obese tend to have lower blood vitamin D levels.
- ❑ People who have undergone gastric bypass surgery, which typically removes the upper part of the small intestine where vitamin D is absorbed.

Conditions resulting from prolonged vitamin D deficiency

Rickets:

A condition in infants and children of soft bones and skeletal deformities caused by failure of bone tissue to harden.



Osteomalacia:

A condition in adults of weak and softened bones that can be reversed with supplementation. This is different than osteoporosis, in which the bones are porous and brittle and the condition is irreversible.



If Vitamin D deficiency continues for long periods, it may result in complications, such as:

- cardiovascular conditions
- autoimmune problems
- neurological diseases
- infections
- pregnancy complications
- certain cancers , including breast, prostate, and colon

Symptoms of vitamin D deficiency

The symptoms of a vitamin D deficiency in adults may include

- tiredness,
- aches,
- pains,
- severe bone or muscle pain or weakness
- stress fractures, especially in your legs, pelvis, and hips

